

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester II)
Scala for Data Science Practical

Practical - 8

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Practical o8

Slice a Breeze matrix to extract a sub-matrix and calculate its row and column Sums.

INPUT

```
import breeze.linalg._

object Demo extends App {
  val m = DenseMatrix((1.0, 2.0, 3.0, 4.0), (5.0, 6.0, 7.0, 8.0), (9.0, 10.0, 11.0,12.0), (13.0, 14.0, 15.0, 16.0))

  val s = m(1 to 2, 1 to 2)
  val r = sum(s(*, :))
  val c = sum(s(:, *))

  println(s"Original Matrix:\n$m")
  println(s"\nSub-Matrix :\n$s")
  println(s"\nRow Sums: $r")
  println(s"Column Sums: $c")
  println("S006 Sejal Deshmukh")
}
```

OUTPUT

```
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...
```

```
Original Matrix:
```

```
1.0  2.0  3.0  4.0  
5.0  6.0  7.0  8.0  
9.0  10.0 11.0 12.0  
13.0 14.0 15.0 16.0
```

```
Sub-Matrix :
```

```
6.0  7.0  
10.0 11.0
```

```
Row Sums: DenseVector(13.0, 21.0)
```

```
Column Sums: Transpose(DenseVector(16.0, 18.0))
```

```
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```

```
Process finished with exit code 0
```